

Worksheet — Incline with a Pulley Simulation

Simulation 04 · use with *Simulation - Incline with pulley.html* · Newton's 2nd law (Blocks 3–4) · goes with card C2★★ and the Block 3 Smart Cart demo

Name: _____ Class: _____ Date: _____

How to use this worksheet

Open **Simulation - Incline with pulley.html** in a browser (it works offline). Set the surfaces, m_1 , m_2 and θ , then press  **Release**. The **Tasks** tab has the same three parts as this sheet; write your **explanations** here.

Part 1 before the lesson (≈ 15 min) · **Part 2** with the Block 3 Smart Cart demo (≈ 10 min) · **Part 3** homework (≈ 30 min).

Use $g = 9.81 \text{ N kg}^{-1}$. Take 'the hanging mass falls, the block moves up the slope' as positive.

Part 1 — Pre-lab: predict, then test

Question	My prediction (before)	What the simulation showed	Physics explanation
1. Card C2★★ (4.0 kg on a smooth 30° slope, 3.0 kg hanging): which way does it move?			
2. While it accelerates, is T less than, equal to or more than m_2g ?			
3. Smooth slope, $m_2 = 2.0 \text{ kg}$ ($m_2g = m_1g \sin 30^\circ$). What happens after release?			
4. Wood slope ($\mu_s = 0.41$), $m_1 = 4.0 \text{ kg}$: which hanging masses keep it at rest?			

5. Draw separate free-body diagrams for the block and for the hanging mass of card C2★★ while they move.

Free-body diagrams (block | hanging mass)

6. Use Newton's 2nd law on each body to calculate the acceleration and the tension for card C2 ★★. Check with the sim.

Working

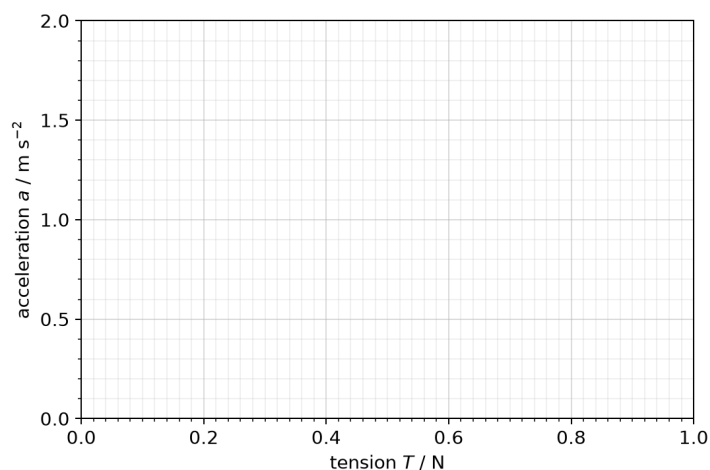
$$a = \text{_____} \text{ m s}^{-2} \quad T = \text{_____} \text{ N}$$

Part 2 — During the lab: Block 3 Smart Cart demo

Sim: Smart Cart wheels on bench, $m_1 = 0.500 \text{ kg}$, $\theta = 0^\circ$. Fill in the sim and the demo's shared data.

m_2 / g	Sim $a / \text{m s}^{-2}$	Sim T / N	Demo $a / \text{m s}^{-2}$	Demo T / N	$m_2 g / \text{N}$
20					
40					
60					
80					
100					

7. Plot a against T (demo or sim data) and draw the best-fit line.



8. Gradient = _____ $\text{kg}^{-1} \Rightarrow$ mass of the cart = _____ kg . x-intercept = _____ N = _____

9. Why is the tension while the cart accelerates smaller than $m_2 g$? Why do we plot a against T , not against $m_2 g$?

Part 3 — Homework: analyse

Wood on bench ($\mu_s = 0.41$, $\mu_d = 0.30$), $m_1 = 1.0$ kg, $\theta = 35^\circ$.

10. Find the smallest and the largest hanging mass that keep the block at rest. Draw the FBD of the block for each case — which way does static friction act?

Free-body diagrams and working

$m_{2,\min} = \underline{\hspace{2cm}}$ kg $m_{2,\max} = \underline{\hspace{2cm}}$ kg

11. $m_2 = 1.5$ kg. Calculate the acceleration and the tension.

Working

12. $m_2 = 0.10$ kg: the block slides **down**. Calculate the size of the acceleration and the tension. Why is T now **greater** than m_2g ?

Working